

Sparse Congruence-Preserving Perturbations of Dyadic Series

What gap statistics can tell us about Erdős Problem #251

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ABSTRACT

We construct convergent dyadic series with both rational and irrational sums while preserving specified congruences and block statistics. Starting from any nonnegative integer coefficients with convergent sum, the nonnegative corrections attain an interval of sums on one set of upper Banach density zero. They eventually satisfy any prescribed bound tending to infinity; coefficient and partial-sum residues are eventually preserved for every fixed modulus. The construction varies the weighted sum while keeping each pair's ordinary total fixed. Applied to prime gaps with correction bound $\log(n+3)$, it preserves the cumulative asymptotic $n \log n$ and gives vanishing total variation error for blocks of length $o(\log \log X)$ sampled on $[X, 2X)$. The cumulative positions need not be prime; the actual prime-series irrationality question remains open.

1 Introduction

Let $p_0 = 2, p_1 = 3, \dots$ be the primes and put $g_n = p_{n+1} - p_n$. Erdős asked whether

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}}$$

is irrational [1, p. 94][2, p. 62] [3, p. 103]. Summation by parts gives $\Pi = 2 + \sum_{n \geq 0} g_n 2^{-(n+1)}$, with convergence justified in Section 4. We study which properties of the gaps survive when the latter sum is changed to a prescribed value.

The point is not merely that one can change the sum. The same family contains rational and irrational values while retaining eventual residues of both the coefficients and their partial sums. Its permitted support and all modulus cutoffs are chosen before the target. With a polylogarithmic correction bound, blocks of length $o(\log \log X)$ have asymptotically the same empirical distributions, and in the prime-gap application the cumulative growth remains $n \log n$. These data therefore do not decide rationality for comparison integer sequences. They do not ensure that the cumulative positions are consecutive primes.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

We use a standard interval-covering argument for series with finite choices. Fridy's generalised-base lemma [5, p. 194] treats prescribed digit bounds and nonincreasing weights. Crmarić and Kovač [17, Lemma 4] give the finite-choice form used here, and Kovač and Tao [6, Lemma 5.1] use analogous intervals of reciprocal choices. We must arrange these choices on a sparse set while preserving cumulative congruences. To do so, we redistribute a fixed correction between two adjacent indices. The simplest useful calculation is

$$\begin{array}{c|cccc} (e_n, e_{n+1}) & (0, 6) & (2, 4) & (4, 2) & (6, 0) \\ e_n + e_{n+1} & 6 & 6 & 6 & 6 \\ 2^{n+2}(e_n 2^{-n-1} + e_{n+1} 2^{-n-2}) & 6 & 8 & 10 & 12. \end{array}$$

Moving 2 from the later index to the earlier one changes the weighted sum while leaving the ordinary sum fixed. If the cumulative correction before the pair is odd, an additional 1 at $n-1$ clears its residue modulo 2. Since the four pairs have the same sum, the choice of pair leaves every later cumulative residue unchanged. We repeat this operation with nested moduli and increasingly separated pairs. The larger separations make the weighted contributions smaller, so we enlarge the range of each pair until the later contributions fill the gaps between consecutive present choices.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, intervals of indices contain integers, and $\log$ is natural unless a base is displayed. A set $S \subseteq \mathbb{N}$ has *upper Banach density* zero if

$$\lim_{H \rightarrow \infty} \sup_{u \in \mathbb{N}} \frac{|S \cap [u, u+H)|}{H} = 0.$$

For integers $X, m \geq 1$, let $\mu_{a,X,m}$ denote the distribution of $(a_n, \dots, a_{n+m-1})$ when n is uniform on $[X, 2X)$. The entries are not rescaled: we sample starting indices, not distinct block values. We use $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, with the supremum over all sets of blocks.

Proposition 1.1 (sparse changes preserving congruences). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \rightarrow \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \rightarrow \mathbb{N}$ satisfying*

$$\text{supp } e \subseteq S, \quad e_n \leq f(n) \text{ eventually}, \quad \sum_{n \geq 0} (a_n + e_n) 2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q , independent of r , such that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n+3))^\varepsilon$ can be made with

$$|S \cap [X, 2X)| = O_\varepsilon \left(\frac{X}{\log \log X} \right), \quad d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X+m)|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log \log X)$, uniformly over target values.

Only e depends on the target; its support may occupy just part of the common set S . Infinitely many changes are essential to changing rationality, since finitely many integer dyadic corrections add a rational number. The size allowance may nevertheless grow as slowly as $\log \log(n + 3)$. It cannot be replaced by a bounded allowance: a modulus exceeding the bound forces e eventually to vanish, after which its constant cumulative sum must be divisible by every positive integer. Nonnegativity then forces $e = 0$. Polynomially growing coefficients satisfy the convergence hypothesis; $a_n = 2^n$ does not.

Section 2 proves the construction and Section 3 transfers the stated prime-gap properties. The rest of the paper returns to the actual primes: integral tail shifts give an exact rationality criterion, whereas two consecutive small differences give a sufficient condition. Finite enclosures certify individual witnesses, not their occurrence arbitrarily late. The [companion's literature discussion](#) contains the further comparisons, including finite subsums and other dyadic coefficient sequences.

2 Proof of Proposition 1.1

Proof. Fixed totals. Each stage must do two things: cancel a cumulative residue and leave a choice of weighted contribution. We use $n_j - 1$ for the cancellation and $n_j, n_j + 1$ for a pair with fixed ordinary sum. Varying that pair then cannot alter the residue to be cancelled at any later stage. Choose an initial index $n_{-1} \geq K$ and centres with $s_j = n_j - n_{j-1} \geq 4$, so the triples do not overlap. Let M_j be positive integers with $M_j \mid M_{j+1}$, and let D_j bound the digit at stage j . We first determine which corrections preserve the congruences; the spacing and digit ranges will then be chosen together.

Let C_j record the ordinary correction before the j th triple. Starting from $C_0 = 0$, prescribe

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \quad C_{j+1} = C_j + c_j + M_j D_j.$$

For integers $0 \leq d_j \leq D_j$, set

$$e_{n_j-1} = c_j, \quad e_{n_j} = M_j d_j, \quad e_{n_j+1} = M_j(D_j - d_j), \quad (1)$$

and put $e_n = 0$ elsewhere. The two sums of a triple are

$$\begin{aligned} \sum_{r=n_j-1}^{n_j+1} e_r &= c_j + M_j D_j, \\ \sum_{r=n_j-1}^{n_j+1} e_r 2^{-r-1} &= c_j 2^{-n_j} + M_j D_j 2^{-n_j-2} + d_j M_j 2^{-n_j-2}. \end{aligned}$$

The ordinary total contains no d_j , so the first identity verifies the meaning of C_j and makes every c_j independent of the digits. The weighted total does contain d_j : transferring M_j from $n_j + 1$ to n_j increases it by

$$w_j = M_j 2^{-n_j-2}.$$

After c_j has been inserted, the cumulative correction is divisible by M_j ; the two entries of the pair preserve this divisibility. Fix $q \mid M_j$. At index n_j both the coefficient correction and the cumulative correction are divisible by q . At every later triple, q divides C_j and M_j , so it also divides c_j and both pair entries. The intervening corrections are zero. Hence

$$q \mid e_n, \quad q \mid \sum_{i < n} e_i \quad (n \geq n_j).$$

The entry c_j need not be divisible by q , which is why it is placed at $n_j - 1$, before the cutoff. The index n_j depends on q and the chosen moduli, but not on the digits or the value to be attained.

Spacing and size. The variable part at stage j takes the values $0, w_j, \dots, D_j w_j$. The covering argument will work if later stages have total range at least one present step, w_j . To retain this range as the centres spread out, we make each diameter a difference of successive dyadic scales. Choose $D_i = 2^{s_i} - 1$, for which

$$D_i w_i = (2^{n_i - n_{i-1}} - 1) M_i 2^{-n_i - 2} = M_i (2^{-n_{i-1} - 2} - 2^{-n_i - 2}).$$

Because $M_i \geq M_j$ for $i > j$, replacing every later modulus by M_j gives the finite estimate

$$\begin{aligned} \sum_{i=j+1}^J D_i w_i &\geq M_j \sum_{i=j+1}^J (2^{-n_{i-1} - 2} - 2^{-n_i - 2}) \\ &= w_j - M_j 2^{-n_J - 2} \quad (J > j). \end{aligned} \tag{2}$$

The digit range $D_j = 2^{s_j} - 1$ therefore gives the required overlap estimate even when the separations increase. Every entry of (1) is at most $M_j 2^{s_j}$, so we must keep this quantity below the prescribed bound at the three indices. We must also have $s_j \rightarrow \infty$ and ensure that every fixed positive integer divides all sufficiently late M_j .

To allow a nonmonotone bound f , put

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

The lower envelope is nondecreasing and tends to infinity, even if f oscillates. Thus a bound imposed at n_{j-1} remains valid at all three later coordinates. The cap by m additionally gives $e_n \leq n$, ensuring summability for every digit choice. Enlarge $n_{-1} \geq K$ until $h(n_{-1}) \geq 32$, and set

$$k_j = \max\{k \geq 2 : k! 2^{k+2} \leq h(n_{j-1})\}, \quad M_j = k_j!, \quad s_j = k_j + 2, \quad n_j = n_{j-1} + s_j.$$

The maximum exists: $k = 2$ is admissible because $2! 2^4 = 32$, and $k! 2^{k+2}$ tends to infinity. Since h is nondecreasing, so is k_j . Moreover $n_j \rightarrow \infty$, hence $h(n_j) \rightarrow \infty$ and $k_j \rightarrow \infty$. It follows that $M_j \mid M_{j+1}$, every fixed q divides all sufficiently late $M_j = k_j!$, and the separations $s_j = k_j + 2$ tend to infinity. This one choice therefore supplies nested moduli and increasing separations while also respecting the size allowance: at every coordinate n of the j th triple,

$$0 \leq e_n \leq M_j 2^{s_j} \leq h(n_{j-1}) \leq \min(f(n), n).$$

This proves the size bound and convergence for every choice of digits.

Let $S = \bigcup_{j \geq 0} \{n_j - 1, n_j, n_j + 1\}$. For any $R \geq 4$, all but finitely many successive centres are at least R apart. An interval of length H therefore contains at most $3(H/R + 2)$ points from these later triples. The earlier triples contribute a fixed finite number, independent of the interval's position. Dividing by H and taking the supremum over positions gives a limiting upper bound $3/R$. Letting $R \rightarrow \infty$ proves upper Banach density zero.

Attaining the interval. The support and congruence requirements are now settled independently of the target. Only the digits remain to be chosen. Define

$$\beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \quad F_j = \sum_{i \geq j} D_i w_i.$$

Here β is the weighted contribution with all digits zero; F_j is the maximum variable contribution from stage j onwards. Both are finite by the bounds just proved, with $\beta > 0$, $F_0 > 0$ and $F_j \rightarrow 0$. The total correction is

$$\sum_n e_n 2^{-n-1} = \beta + \sum_j d_j w_j.$$

Taking $J \rightarrow \infty$ in (2) now gives

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \quad (3)$$

Every variable sum from stage j onwards lies in $[0, F_j]$; we must still show that each point is attained. Choosing digit d leaves the next remainder in $[0, F_{j+1}]$ exactly when the current remainder belongs to

$$[dw_j, dw_j + F_{j+1}], \quad d = 0, \dots, D_j.$$

These intervals meet or overlap by (3), and their union is $[0, F_j]$ because $F_j = D_j w_j + F_{j+1}$. Figure 1 shows the two calculations. Given $x \in [0, F_0]$, start with $\rho_0 = x$. Having chosen the first j digits, choose d_j so that

$$\rho_{j+1} = \rho_j - d_j w_j \in [0, F_{j+1}].$$

The covering just proved makes this possible at every stage. Since $0 \leq \rho_j \leq F_j$ and $F_j \rightarrow 0$, we obtain

$$x - \sum_{i=0}^j d_i w_i = \rho_{j+1} \rightarrow 0.$$

Thus every $x \in [0, F_0]$ is attained, and we may take $I = (A + \beta, A + \beta + F_0)$. The target $r \in I$ determines only the digits, through $x = r - A - \beta$; it changes neither S , I nor the cutoff $N_q = n_j$ chosen using $q \mid M_j$. This last selection is the standard finite-choice covering argument of [17, Lemma 4]. The construction above supplies its summability and overlap hypotheses while imposing the congruences and requiring the support to lie in S .

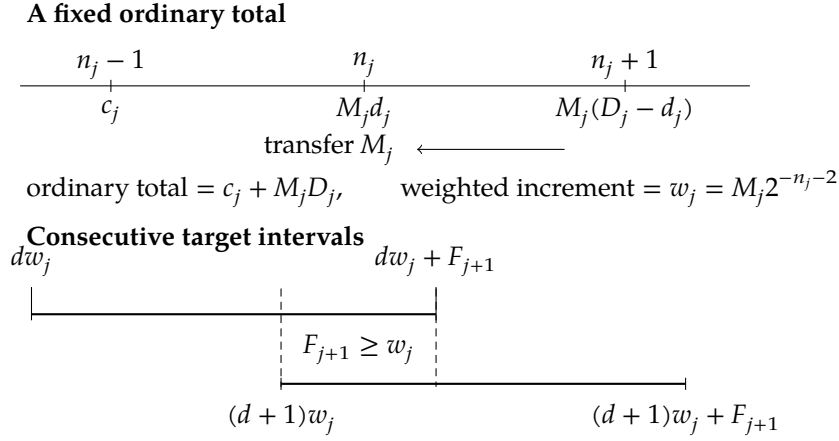


Figure 1: Redistributing M_j between adjacent entries keeps their ordinary sum fixed and increases the dyadic sum by w_j . The candidate target intervals in the lower panel cover without gaps when $F_{j+1} \geq w_j$. The sketch shows strict overlap; the proof also allows equality and establishes attainment by keeping the remainder in $[0, F_j]$ as $F_j \rightarrow 0$.

The logarithmic scale. For $f(n) = (\log(n+3))^\varepsilon$, we split the available size between the modulus M_j and the spacing cost 2^{s_j} . Keep the same triples and $D_j = 2^{s_j} - 1$, and choose

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor,$$

$$k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\},$$

with $M_j = k_j!$ and $n_j = n_{j-1} + s_j$. Start far enough out that $s_j \geq 4$ and all choices exist. The exponents $\varepsilon/4$ and $\varepsilon/2$ add to $3\varepsilon/4 < \varepsilon$, so the corrections satisfy the prescribed bound:

$$M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4} \leq (\log(n+3))^\varepsilon$$

at each coordinate n of the triple. After enlarging the initial index, this is also at most n . The moduli remain nested, and every fixed positive integer divides all sufficiently late moduli. Summability, the congruence induction and (2) therefore apply without change. Since $s_j \asymp_\varepsilon \log \log n_j$, the relevant centres near $[X, 2X)$ are separated by $\gg_\varepsilon \log \log X$. Each contributes at most three support points, giving

$$|S \cap [X, 2X)| = O_\varepsilon(X / \log \log X).$$

Couple the two blocks by choosing the same uniform starting index $n \in [X, 2X)$. They agree whenever the sampled block avoids S . A changed coordinate t can occur only in blocks starting at $n \in [t - m + 1, t]$, at most m starts. Only $t \in S \cap [X, 2X + m)$ can occur in a sampled block. The union bound consequently gives

$$\begin{aligned} d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) &\leq \mathbb{P}(e_{n+i} \neq 0 \text{ for some } 0 \leq i < m) \\ &\leq \frac{m|S \cap [X, 2X + m)|}{X}. \end{aligned}$$

For $m \leq X$, two dyadic bands cover $[X, 2X + m)$ and give the bound $O_\varepsilon(m / \log \log X)$. In particular it tends to zero when $m = o(\log \log X)$, uniformly over the values attained, since the same set S contains every correction support. $\square$

No limiting block distribution is assumed: the coupling compares the two empirical laws at each X . It also bounds the change in the mean of any index-dependent test of absolute value at most B_0 by $2B_0 m |S \cap [X, 2X + m)| / X$; for a block-only test the bound is $2B_0 d_{\text{TV}}$. These are absolute errors, not relative estimates for rare events or comparisons of complete infinite tails. The scale $o(\log \log X)$ is sufficient here, not proved optimal. The companion's [Section 2](#) also distinguishes upper Banach density from ordinary density.

3 Application to prime gaps

The elementary bound $p_n \leq 1250(n+1)^4$ suffices for convergence of the prime and gap series. We use the central-binomial argument of Erdős [19, pp. 194–196]; [Appendix A of the companion](#) derives this bound, including the prime-power estimate. This supplies the convergence hypothesis needed to apply [Proposition 1.1](#) to the prime gaps.

Corollary 3.1 (a rational sum with the stated prime-gap statistics). *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \leq 1$, there is $b : \mathbb{N} \rightarrow \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \geq g_n$, and $b_n - g_n \leq (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The empirical distributions of unnormalised blocks have total variation distance tending to zero for lengths $o(\log \log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i < n} b_i$,

$$0 \leq P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log \log n}\right), \quad P_n \sim n \log n.$$

These positions are not asserted to be prime.

Lean proves this for every $\varepsilon > 0$, assuming Schläge-Puchta's Lemma 4 [18] only for the nonconcentration bound and the prime number theorem only for $P_n \sim n \log n$.

Proof. Apply [Proposition 1.1](#) to $a = g$, choose a rational $r \in I$, and fix the resulting sequence $b = g + e$. The correction, congruence and growing-block assertions follow at once. It remains to transfer polynomial nonconcentration and cumulative growth. For fixed k and nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$, a new zero requires a block meeting S , so

$$\begin{aligned} & |\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| \\ & \leq |\{n < N : F(g_n, \dots, g_{n+k}) = 0\}| + \sum_{i=0}^k |S \cap [i, N+i]|. \end{aligned}$$

The first count is $o(N)$ by Schlage-Puchta's Lemma 4 [18]; the finitely many support counts are $o(N)$ because S has density zero. Only changed blocks are counted, so the values of F need not be bounded. Since b was chosen before F, k , the same sequence satisfies every fixed-polynomial condition.

To estimate $P_n - p_n = \sum_{i < n} e_i$, first observe that $|S \cap [0, n)| = O_\varepsilon(n / \log \log n)$. Indeed, the indices below $\sqrt{n}$ contribute at most $\sqrt{n}$ points, and on $[\sqrt{n}, n)$ we sum the dyadic support bounds, whose lengths add to $O(n)$ and whose $\log \log X$ are comparable to $\log \log n$. Multiplying by the pointwise correction bound gives the stated error. For $0 < \varepsilon \leq 1$ this is $o(n \log n)$, so the prime number theorem [12, Chapter 6] gives $P_n \sim n \log n$. Finally, summing nonnegative terms in the opposite order, we obtain

$$\sum_{n \geq 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i \geq 0} b_i \sum_{n > i} 2^{-n-1} = 2 + \sum_{i \geq 0} \frac{b_i}{2^{i+1}} = 2 + r \in \mathbb{Q}.$$

□

For $\varepsilon > 1$ we may use exponent 1, which gives smaller corrections and the error $O(n \log(n+3) / \log \log n)$. Thus the conclusions extend to every $\varepsilon > 0$. For each fixed $y \geq 2$, eventual congruence modulo $y!$ also gives $\gcd(P_n, y!) = 1$ once $p_n > y$. The cutoff depends on y : this does not prove that P_n avoids every prime up to $\sqrt{P_n}$. Even prime-valued positions would not ensure that no primes are omitted.

Land's draft [8, Theorem 2] proves conditional irrationality under a uniform Hardy–Littlewood hypothesis of the kind formulated by Kuperberg [7, Conjecture 1.3]. Ringer's draft [16] obtains conditional normality. Their quantitative prime-pattern hypotheses are stronger than the absolute block comparison above and are not assumed here. The companion's Section 3 compares the hypotheses and sampling conventions.

4 The prime series and its actual tails

The construction shows what the specified coefficient data fail to determine. We now ask what would determine rationality for the actual primes. Write $G = \sum_{n \geq 0} g_n 2^{-(n+1)}$. The polynomial bound from Section 3 proves convergence and makes the endpoint in finite summation by parts vanish.

Theorem 4.1 (prime-to-gap identity). *The actual prime and gap series satisfy $\Pi = 2 + G$. Their complete tails*

$$T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ and $T_{N+1} = 2T_N - g_{N+1}$. Thus Π , G and T_0 have the same rationality status.

Proof. Finite summation by parts gives

$$\sum_{i=0}^n \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Since $p_n = O((n+1)^4)$, the endpoint tends to zero and we obtain $\Pi = 2 + G$. We obtain the tail recurrence by splitting off the first term of T_N ; removing $g_0 = 1$ from G gives $T_0 = 2G - 1$. □

Tao noted this reduction in the problem's forum discussion [15, 7 October 2025].

Corollary 4.2 (exact irrationality reformulation). *The prime-value dyadic series is irrational if and only if the prime-gap dyadic series is. Both series converge by the polynomial bound proved above; neither side is proved irrational.*

Proof. Adding the rational number 2 preserves rationality and irrationality. $\square$

The index N is the last omitted coefficient: the first term of T_N is $g_{N+1}/2$. These are rescaled complete tails, not the unscaled remainders of G . The recurrence alone does not identify them, since adding $C2^N$ preserves it. The next lemma excludes this extra term.

Lemma 4.3 (the boundary condition for the tail recurrence). *Let $\sum_{j \geq 1} |a_j|2^{-j} < \infty$ and $U_{N+1} = 2U_N - a_{N+1}$. Then $U_N = \sum_{j \geq 1} a_{N+j}2^{-j}$ for every N if and only if $2^{-N}U_N \rightarrow 0$.*

Proof. Iteration gives $2^{-N}U_N = U_0 - \sum_{j=1}^N a_j2^{-j}$. The limit is zero exactly when U_0 is the full series. Subtracting its first N terms and multiplying by 2^N then gives the tail formula. Conversely the formula implies $2^{-N}U_N = \sum_{k > N} a_k2^{-k} \rightarrow 0$ by absolute convergence. $\square$

4.1 Integral shifts

For the recurrence $U_{N+1} = 2U_N - a_{N+1}$ with $a_n \in \mathbb{Z}$, write $D_h(N) = U_{N+h} - U_N$. Modulo integers, each step is doubling: U_N has the fractional part of $2^N U_0$. An integral shift means that a fractional part has returned to an earlier value. Such a return makes $2^N(2^h - 1)U_0$ an integer and forces rationality when $h > 0$. Conversely, doubling a rational number eventually gives periodic fractional parts. The next theorem records the denominator calculation and the exact quantifiers.

Theorem 4.4 (exact rationality classification). *For a real integer-coefficient recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N . More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

$$\text{den}(U_N) = 2^{\max(s-N, 0)} d, \quad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives

$$U_N - 2^N U_0 \in \mathbb{Z}, \quad D_h(N) - 2^N(2^h - 1)U_0 \in \mathbb{Z}.$$

An integral shift with $h \geq 1$ makes the nonzero integer multiple $2^N(2^h - 1)U_0$ integral, and therefore forces $U_0 \in \mathbb{Q}$. Conversely, write $U_0 = u/(2^s d)$ in lowest terms. Multiplication by 2^N cancels exactly $\min(N, s)$ powers of two from the denominator. The remaining factor $2^h - 1$ is odd, so the second relation is integral exactly when $N \geq s$ and $d \mid 2^h - 1$. For $d > 1$, Euler's congruence provides such a shift $h = \varphi(d)$, and for $d = 1$ we take $h = 1$. This shift works for every $N \geq s$. Negating the pointwise and eventual criteria gives the two assertions about irrationality. $\square$

The two preceding results have different roles. Lemma 4.3 identifies the complete tails; the classification applies to every integer-coefficient recurrence. Its periodicity concerns fractional parts, not necessarily full tails. For instance, the positive even coefficients 2, 4, 2, 4, ... starting at $n = 1$ give tails 8/3, 10/3 alternately. Every shift of length 1 is nonintegral and every shift of length 2 is zero. This explains why a single shift length is insufficient.

5 Small tail differences and finite tests

Fix $h \geq 1$ and put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$. Both gap indices are positive, so neither involves the exceptional odd gap $g_0 = 1$; hence δ_N is even. Two integers of absolute value less than one must both vanish, forcing $\delta_N = 0$. Evenness yields the sharper signed test below.

Proposition 5.1 (two consecutive differences of absolute value less than one). *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

$$\delta = 2s, \quad \frac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both D and D' are nonintegral.

Proof. The bounds on D, D' imply $-3 < \delta < 3$, so a nonzero even δ equals $2s$ with $s \in \{-1, 1\}$. Substituting in $D' = 2D - 2s$ gives $1/2 < sD < 1$. Conversely this interval gives $sD' = 2sD - 2 \in (-1, 0)$. $\square$

The endpoints cannot be included: when $\delta = 2$, $D = 1/2$ gives $D' = -1$, and $D = 1$ already violates $|D| < 1$. The zero triple $D = D' = \delta = 0$ shows why the mismatch is also required.

Problem 5.2 (two small tail differences at arbitrarily large indices). For every integer $h \geq 1$ and every cutoff N_0 , exhibit $N \geq N_0$ with

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \quad (4)$$

This is a sufficient condition, not another equivalence: the exact classification requires nonintegrality, not smallness. A solution gives the shifts required by Theorem 4.4. All three conditions must hold at the same index; separate infinite sets of witnesses do not suffice.

By Proposition 5.1, such an index must have $\delta_N = \pm 2$. Schlage-Puchta's lemma [18, Lemma 4], applied to $x_h - x_0 \pm 2$, shows that these eligible indices already have density zero. Thus the required witnesses concern a rare event whose relative frequency is not controlled by the absolute block comparison in Proposition 1.1.

5.1 Finite separation from the integers

To certify a nonintegral difference from finitely many gaps, we need an enclosure disjoint from the integers; an enclosure meeting an integer is inconclusive. Choose $M(n) \geq g_n$

with $\sum_n M(n)2^{-n} < \infty$, and write

$$S_{h,N,L} = \sum_{j=1}^L (g_{N+h+j} - g_{N+j})2^{-j}, \quad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j))2^{-j}.$$

Splitting the complete tails after L terms gives

$$D_N = S_{h,N,L} + 2^{-L}D_{N+L}, \quad |D_N - S_{h,N,L}| \leq R_{h,N,L}(M).$$

The uncomputed part is another complete tail difference, with the same shift h . The criterion below uses only the displayed bound; a general real majorant need not make that bound computable.

For the actual prime gaps, one computable choice is $M(n) = 1250(n+2)^4$. For $P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150$, the identity $2P(x) = (x+1)^4 + P(x+1)$ telescopes to $\sum_{j=1}^J (x+j)^4 2^{-j} = P(x) - 2^{-J}P(x+J)$. The endpoint vanishes as $J \rightarrow \infty$, giving

$$R_{h,N,L}(M) = 1250 \cdot 2^{-L} (P(N+h+L+2) + P(N+L+2)).$$

Proposition 5.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\text{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \tag{5}$$

then Π is irrational.

Proof. The enclosure gives $\text{dist}(D_N, \mathbb{Z}) \geq \text{dist}(S_{h,N,L}, \mathbb{Z}) - R_{h,N,L}(M) > 0$. Thus each fixed positive shift is nonintegral at arbitrarily large indices, and Theorem 4.4 applies. $\square$

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \rightarrow 0$, nonintegrality of D is equivalent to $\text{dist}(S_L, \mathbb{Z}) > R_L$ for some L . Indeed, for $\delta = \text{dist}(D, \mathbb{Z}) > 0$, take $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Unrestricted depth therefore gives an exact reformulation, not an additional analytic estimate. At a prescribed logarithmic depth one must also control the margin from the integers as N varies; a shrinking remainder alone does not do so. The companion, Appendix D.2, gives an explicit example distinguishing refinement at a fixed value from enclosures of changing values.

A nonintegral value need not lie in either signed interval of Proposition 5.1. To certify (4), suppose $\delta_N = 2s$, with $s \in \{-1, 1\}$, and enclose D_N using integers A, Q, B with $Q > 0$, $B \geq 0$ and $|QD_N - A| \leq B$. Then $sD_N \in [(sA - B)/Q, (sA + B)/Q]$. Placing its lower endpoint above $1/2$ and its upper endpoint below 1 gives, respectively,

$$2sA - Q > 2B, \quad Q - sA > B.$$

Reducing A modulo Q would preserve its distance to $Q\mathbb{Z}$ but lose the enclosure's location. We retain the signed numerator and use the sign s fixed by the gap difference. At $h = 1, N = 2, L = 40$, one has $\delta_2 = g_4 - g_3 = -2$ and

$$Q = 2^{40}, \quad A = -662838684750, \quad B = 11764181250, \quad s = -1.$$

Here $A/Q = S_{1,2,40}$ and $B/Q = R_{1,2,40}(M)$, and the two strict margins are 202637379224 and 424908761776. The [companion's integer certificates](#) give the prime-index calculation and remainder derivation. This finite pair leaves the quantifiers in Problem 5.2 unresolved.

6 Further consequences and questions

We can place the interval in Proposition 1.1 inside $(A, A + \eta)$ for any $\eta > 0$. Choose $K' \geq K$ such that $(K' + 1)2^{-K'} < \eta$, and start the construction after K' . The bound $e_n \leq n$ from the proof gives

$$0 < \sum_n e_n 2^{-(n+1)} \leq (K' + 1)2^{-K'} < \eta.$$

Thus rational and irrational targets remain available arbitrarily close to A , with the prescribed prefix unchanged. The estimate controls the dyadic change, not the ordinary total.

6.1 Nonperiodic coefficients

Rationality forces eventual periodicity of binary digits, not of an arbitrary integer coefficient sequence. Carrying changes the sequence to which that criterion applies. For example,

$$\sum_{i=0}^{n-1} \frac{i-1}{2^{i+1}} = -\frac{n}{2^n} \rightarrow 0.$$

Its coefficients are unbounded and its first coefficient is -1 . We obtain a positive even example by taking $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n+4)^2$. Direct substitution gives $U_{n+1} = 2U_n - c_{n+1}$, while the polynomial growth gives $2^{-n}U_n \rightarrow 0$. Lemma 4.3 therefore identifies U as the complete tail and yields

$$\sum_{j \geq 1} c_j 2^{-j} = 32, \quad \sum_{n \geq 0} c_n 2^{-(n+1)} = 18.$$

The coefficients increase strictly and $c_{n+1} - c_n = 4n + 10$ is never ± 2 . The [companion's Section 8](#) gives further examples, including a bounded nonperiodic sequence in Appendix D. None of these examples constructs consecutive primes.

Proposition 6.1 (prime gaps do not become periodic). *For every positive h , the actual consecutive-prime-gap sequence is not eventually periodic with period h .*

Proof. If the gaps were eventually periodic, they would be bounded. For each $m \geq 2$, however, the $m-1$ integers $m!+2, \dots, m!+m$ are composite. The consecutive primes on either side therefore have a gap of at least m , contradicting boundedness. $\square$

The results on bounded prime gaps [13], clusters of each fixed size [11, 20], and extreme large gaps [4] do not give the simultaneous inequalities in (4). This remains so with Polymath's bound $\liminf_n g_n \leq 246$ [20, Theorem 1.4(i)]. For fixed h and $\varepsilon > 0$, taking $L = \lceil (4 + \varepsilon) \log_2(N + 2) \rceil$ in the polynomial remainder gives an error $O_{h,\varepsilon}(N^{-\varepsilon})$.

The finite sum must still clear the relevant boundary by more than this error to supply a witness for Problem 5.2. The perturbation theorem rules out no argument using additional primality information or quantitative rare-event estimates.

7 Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The linked Lean declarations use Lean 4 [9] and mathlib [10]. The formal proof of Proposition 1.1 changes one coordinate at each stage, whereas the proof here uses triples. Both choose the set containing the supports, the interval and the congruence cutoffs before the value of the sum. For Corollary 3.1, the formal statement covers every $\varepsilon > 0$ and assumes Schlage-Puchta’s lemma only for nonconcentration and the prime number theorem only for the asymptotic $P_n \sim n \log n$. The source links record formal support for the linked statements, rather than verification of the different printed proofs. The [companion’s Appendix G](#) retains the complete source index at this note’s original pin, including the finite telescoping identities and their nonperiodicity consequences. Its [Section 3](#) retains the broader subsum and automatic-sequence comparisons.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Proposition 1.1	Lean · Comparator	Corollary 3.1	Lean [†]
Theorem 4.1	Lean · Comparator	Corollary 4.2	Lean · Comparator
Lemma 4.3	Lean · Comparator	Theorem 4.4	Lean · Comparator
Proposition 5.1	Lean · Comparator	Proposition 5.3	Lean · Comparator
Proposition 6.1	Lean · Comparator		